

Project FORESIGHT

Demand & Inventory Intelligence — Executive readout

What to reorder, what to clear, and what to leave alone this month — with the rupee value of each decision and an honest account of how much to trust the forecast behind it.

For: Head of Operations · Finance lead · Merchandiser | Prepared by: Eswar Mahalingam, Data Scientist | Forecast origin 2025-11-30 · 8-week horizon | Live: <https://foresight-northbay.netlify.app>

The one-slide answer

₹47.9 L of sales is about to be lost and ₹1.48 Cr of cash is sitting in stock that will not sell for months. Both are fixable this week.

Sales at risk from stockouts

₹47.9 L

8 SKUs will run out before a replenishment can land — **reorder now**

Capital locked in overstock

₹1.48 Cr

8 SKUs hold 15–70 weeks of cover — **promote or clear**

Healthy — leave alone

34 / 50

No action; free the team's attention for the 16 that matter

How much to trust it

8.8 %

forecast error (WAPE) on unseen weeks vs 11.2 % for "same as last year" — 22 % better

Do this week: raise purchase orders for the 8 reorder SKUs (quantities on slide 4, 4,586 units total). SKU012 alone has 2 days of stock left against a 12-day lead time.

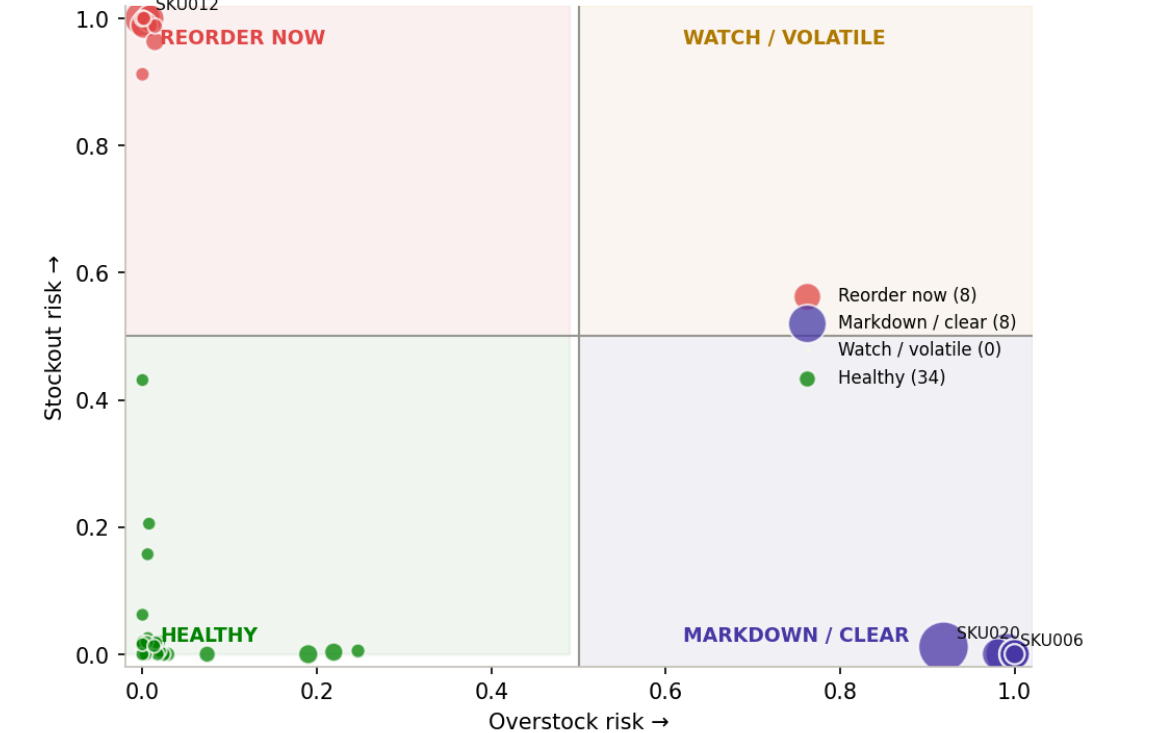
Do this month: run a clearance promotion on the 8 overstock SKUs (slide 5). Exclude SKU020, SKU011 until Finance confirms their cost/price — they are recorded as selling below cost.

All figures from the FORESIGHT pipeline run on NorthBay's extracts (Jan-2024 → Dec-2025); latest stock position = 1-Dec-2025 snapshot. Rupee values use list price for lost sales and unit cost for locked capital.

Every SKU on one grid — the team triages 50 products in 30 seconds

Up = likely to stock out during the lead time. Right = holding far more than 8 weeks of demand. Bubble = rupees at stake.

Decisioning view — every SKU on the stockout vs overstock grid (bubble = ₹ at stake)



Quadrant	SKUs	Action
Reorder now	8	Raise replenishment before stock runs out
Markdown / clear	8	Promote or discount to free cash
Watch / volatile	0	Investigate erratic demand manually
Healthy	34	Nothing — leave as is

No black box. Stockout risk = the probability that demand over the supplier lead time exceeds what is on the shelf, using the forecast and its measured error. Overstock = weeks of cover beyond the 8-week horizon. Anyone can recompute a score with a calculator — and the dashboard's what-if slider lets the team test "what if the lead time slips to 21 days?" instantly.

Reorder now — 8 SKUs, ₹47.9 L of sales at risk in the lead time alone

Order quantity covers demand through the lead time plus a 4-week review cycle, plus safety stock, minus what is already on hand and on order.

SKU	Product	Category	On hand	On order	Lead time (d)	Days of stock left	Fcst / wk	Order qty	Sales at risk
SKU012	Product 012	Home Decor	47	60	12	2	171.5	919	₹20.5 L
SKU017	Product 017	Home Decor	95	26	12	8	88.3	506	₹10.8 L
SKU031	Product 031	Furniture	93	40	13	5	124.5	617	₹9.9 L
SKU042	Product 042	Home Decor	90	120	6	4	148.3	559	₹3.2 L
SKU040	Product 040	Storage	67	29	9	5	97.5	512	₹1.3 L
SKU034	Product 034	Lighting	34	103	5	2	130.2	536	₹87,590
SKU010	Product 010	Storage	83	70	13	5	109.5	519	₹75,057
SKU023	Product 023	Kitchen	130	79	12	9	98.1	418	₹38,906

Why the number looks small next to the revenue: ₹47.9 L is only what is lost *before the next order could arrive*. Left unordered for the full 8 weeks these 8 SKUs represent ₹4.28 Cr of demand — they are among NorthBay's best sellers (e.g. SKU012, SKU017).

Markdown / clear — 8 SKUs hold ₹1.37 Cr of the ₹1.48 Cr locked beyond the 8-week horizon

Excess units = position minus 8 weeks of forecast demand; locked capital values them at unit cost.

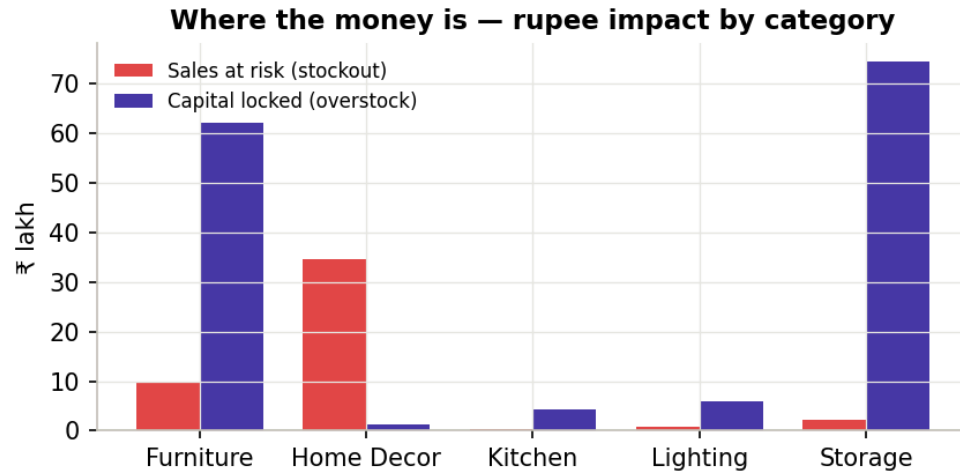
SKU	Product	Category	On hand	On order	Fcst / wk	Weeks of cover	Excess units	Locked capital	Margin
SKU020	Product 020	Storage	1002	353	88.5	15.3	647	₹43.3 L	below cost
SKU006	Product 006	Furniture	359	485	44.2	19.1	491	₹29.5 L	ok
SKU011	Product 011	Furniture	753	269	17.5	58.3	882	₹15.2 L	below cost
SKU036	Product 036	Furniture	383	135	32.0	16.2	262	₹14.2 L	ok
SKU025	Product 025	Storage	1124	61	17.1	69.1	1048	₹14.0 L	ok
SKU015	Product 015	Storage	187	245	23.5	18.4	244	₹13.3 L	ok
SKU004	Product 004	Lighting	259	254	29.0	17.7	281	₹4.1 L	ok
SKU050	Product 050	Storage	328	525	38.9	21.9	542	₹3.8 L	ok

Quick win: SKU020 and SKU006 together hold ₹72.9 L. Also **cancel or defer the open orders** on these lines (2,327 units on order against products already at 15–70 weeks of cover).

Caution: SKU020, SKU011 are recorded as selling below cost. Discounting them deepens the loss. Fix the master data (or the price) first.

Where the money is, by category

Home Decor carries the stockout exposure; Storage and Furniture carry the overstock. Two different conversations with two different buyers.

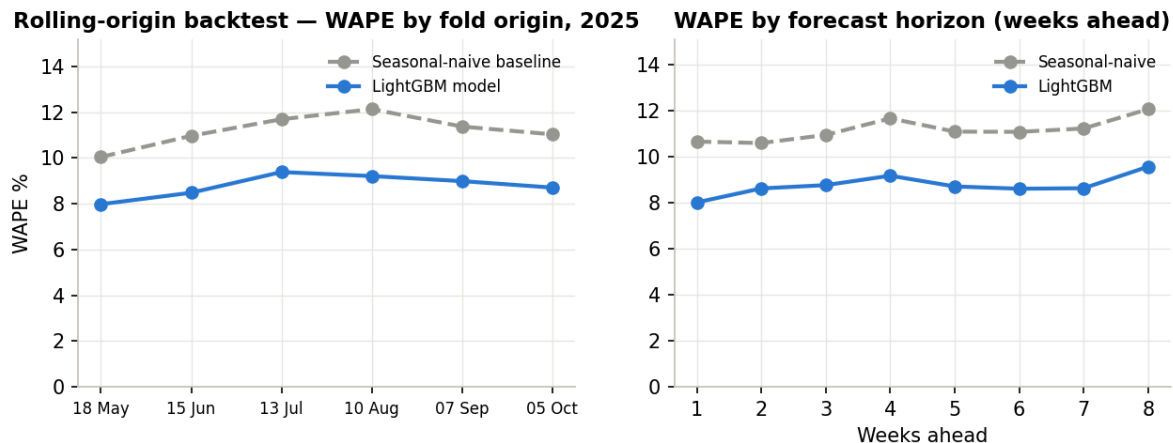


Category	SKUs	Sales at risk	Locked capital
Furniture	10	₹9.9 L	₹62.0 L
Home Decor	10	₹34.6 L	₹1.2 L
Kitchen	10	₹38,906	₹4.2 L
Lighting	10	₹87,590	₹5.8 L
Storage	10	₹2.1 L	₹74.4 L

Pattern behind the numbers: demand peaks in March and troughs in October (a 52 % swing) and the same happens every year. Promotions add 38 %, weekends 25 %. A flat monthly order quantity cannot follow that curve — which is exactly how both problems arise at once.

How much to trust the forecast — tested the honest way

Retrained at six past dates using only earlier data, then scored on the following 8 weeks it had never seen. It beat "same week last year" every time.



WAPE = total forecast miss ÷ total units sold. 8.8 % means roughly 9 units wrong per 100 sold.

FORESIGHT model

8.8 %

bias +0.3 % — not systematically high or low

"Same as last year" baseline

11.2 %

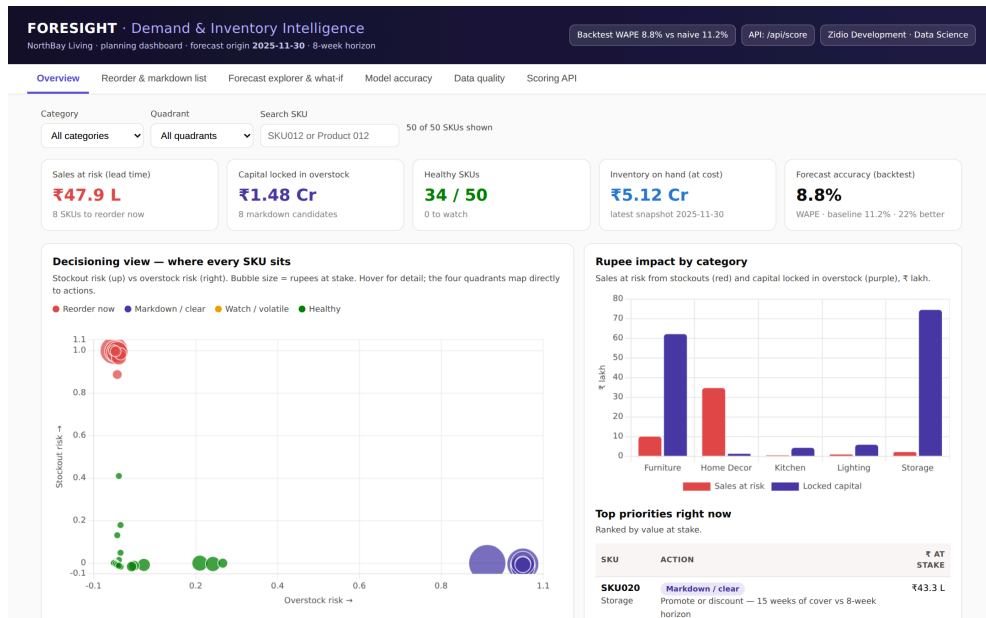
model is 22 % more accurate; wins on 49/50 SKUs

December 2025 (true holdout)

9.3 %

vs 12.4 % baseline · the 80 % range contained 80 % of actuals

What the team gets, and how to run it without a data scientist



Planning dashboard — <https://foresight-northbay.netlify.app>

- **Planning dashboard** (live URL above): filters by category / SKU / quadrant, reorder and markdown lists, per-SKU forecast with uncertainty band, what-if slider for lead time and stock, one-click CSV export.
- **Scoring service**: <https://foresight-northbay.netlify.app/api/score?sku=SKU012> returns the forecast and risk for any SKU or batch — pluggable into NorthBay's own tools.
- **Reproducible pipeline**: drop next month's four extracts in data/raw/, run one command, every number refreshes in about a minute.
- **Monthly rhythm**: refresh → review the two lists → raise POs / schedule promos → check last month's error on the Model tab.

Ask of NorthBay: (1) confirm cost/price on the 16 below-cost SKUs; (2) reconcile the 150 inventory SKUs with no sales (₹17.49 Cr of stock); (3) share the promotion calendar 8 weeks ahead — it is the single biggest accuracy lever.

Decisions requested today

#	Decision	Owner	Value	When
1	Approve purchase orders for the 8 "reorder now" SKUs (4,586 units, quantities on slide 4)	Head of Operations	Protects ₹47.9 L in the lead time; ₹4.28 Cr over 8 weeks	This week
2	Clearance promotion on the 6 overstock SKUs with healthy margin; defer/cancel their open orders	Merchandiser	Releases up to ₹78.9 L of cash	This month
3	Verify cost & list price on the 16 below-cost SKUs before any discounting	Finance lead	Avoids deepening a loss on 17 % of revenue	Before decision 2
4	Reconcile SKU051–SKU200 inventory records (no sales in 2 years)	Operations + Finance	₹17.49 Cr of stock unaccounted for in demand terms	This quarter
5	Adopt the monthly refresh cadence and share the promo calendar 8 weeks ahead	Head of Operations	Keeps forecast error near 9 % instead of drifting	Ongoing

Bottom line: NorthBay does not have a demand problem — demand is predictable to within about 9 %. It has a *timing* problem: fast sellers are ordered too late and slow sellers too often. FORESIGHT turns that into two short lists every month. Deliver it like a consultant, defend it like a scientist.